DATA STRUCTURES

HOMEWORK ASSIGNMENT NO:1

1.SUM OF TWO MATRICES

```
#include<stdio.h>
int main()
{
    int matrix1[10][10],matrix2[10][10],sum[10][10];
    int i,j,rows,cols;
    printf("Enter the number of rows:");
    scanf("%d",&rows);
    printf("Enter the number of columns:");
    scanf("%d",&cols);
    printf("Enter the values of matrix1:");
    for(i=0;i<rows;i++)
    {
        for(j=0;j<cols;j++)
        {
            scanf("%d",&matrix1[i][j]);
        }
    }
    printf("Enter the values of matrix2:");
    for(i=0;i<rows;i++)
    {
        for(j=0;j<cols;j++)
        {
            scanf("%d",&matrix2[i][j]);
        }
    }
    printf("Sum of the matrices:\n");
```

```
    for(i=0;i<rows;i++)
    {
        for(j=0;j<cols;j++)
        {
            sum[i][j]=matrix1[i][j]+matrix2[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
Enter the number of rows:2
Enter the number of columns:2
Enter the values of matrix1:1
2
3
4
Enter the values of matrix2:5
6
7
8
Sum of the matrices:
6 8
10 12
```

2.C PROGRAM TO READ N NUMBER OF VALUES IN AN ARRAY AND DISPLAY THEM IN REVERSE ORDER.

```c
#include<stdio.h>
int main()
{
   int arr[100],n,i;
   printf("Enter the number of values in array:");
   scanf("%d",&n);
   printf("Enter the array elements:");
   for(i=0;i<n;i++)
   {
   scanf("%d",&arr[i]);
   }
   printf("\nArray elements:\n");
   for(i=0;i<n;i++)
   {
      printf("%d ",arr[i]);
   }
   printf("\nArray elemnts in reverse order\n");
   for(i=n-1;i>=0;i--)
   {
      printf("%d ",arr[i]);
   }
   return 0;
}
```

OUTPUT:

Enter the number of values in array:5

Enter the array elements:8

9

5

4

6

Array elements:

8 9 5 4 6

Array elemnts in reverse order

6 4 5 9 8

3.COUNT THE TOTAL DUPLICATE ELEMENTS IN AN ARRAY

```
#include<stdio.h>
int main()
{
   int arr[50],dup[50],i,j,k,n,duplicate=0,found=1;
   printf("Enter number of elements in the array:");
   scanf("%d",&n);
   printf("Enter array elements:");
   for(i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }
   for(i=0;i<n;i++)
   {
      for(j=i+1;j<n;j++)
      {
         if(arr[i]==arr[j])
         {
            found=0;
            for(k=0;k<duplicate;k++)
            {
               if(dup[k]==arr[i])
               {
```

```
                found=1;
                break;
            }
        }
        if(found==0)
        {
            dup[duplicate]=arr[i];
            duplicate++;
        }
      }
    }
  }
  printf("Duplicate elements:\n");
  for(i=0;i<duplicate;i++)
  {
    printf("%d ",dup[i]);
  }
  printf("\nTotal number of duplicate elements:%d",duplicate);
  return 0;
}
```

OUTPUT:

```
Enter number of elements in the array:5
Enter array elements:66
77
78
78
90
Duplicate elements:
78
```

Total number of duplicate elements:1

4.PRINT UNIQUE ELEMENTS IN AN ARRAY

```
#include<stdio.h>
int main()
{
    int count;
    int arr[50],i,unique[50],u=0,n,j;
    printf("Enter number of elements in the array:");
    scanf("%d",&n);
    printf("Enter array elements:");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(i=0;i<n;i++)
    {
        count=0;
        for(j=0;j<n;j++)
        {
            if(i!=j && arr[i]==arr[j])
            {
                count++;
            }
        }
        if(count==0)
        {
            unique[u]=arr[i];
            u++;
        }
```

```
    }
    printf("Unique elements:\n");
    for(i=0;i<u;i++)
    {
        printf("%d ",unique[i]);
    }
    printf("\nTotal number of unique elements:%d",u);
    return 0;
}
```

OUTPUT:

```
Enter number of elements in the array:6
Enter array elements:5
4
7
8
2
5
Unique elements:
4 7 8 2
Total number of unique elements:4
```

5.INSERT IN SORTED ARRAY

```
#include<stdio.h>

int main()
{
    int arr[50], n, i, value;

    printf("Enter the number of elements: ");
    scanf("%d",&n);
```

```
    printf("Enter the sorted array elements:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }

    printf("Enter the value to insert: ");
    scanf("%d",&value);

    i = n - 1;

    while(i >= 0 && arr[i] > value)
    {
        arr[i+1] = arr[i];
        i--;
    }

    arr[i+1] = value;
    n++;

    printf("Array after insertion:\n");
    for(i=0;i<n;i++)
    {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

OUTPUT:

Enter the number of elements: 5

Enter the sorted array elements:

3

5

7

9

10

Enter the value to insert: 6

Array after insertion:

3 5 6 7 9 10

6.INSERT IN UNSORTED ARRAY

```
#include<stdio.h>
int main()
{
    int arr[40],n,pos,i,value;
    printf("Enter the number of elements in the array:");
    scanf("%d",&n);
    printf("Enter the value to be inserted:");
    scanf("%d",&value);
    printf("Enter the position to insert:");
    scanf("%d",&pos);
    printf("Enter the array elements:");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(i=n-1;i>=pos-1;i--)
    {
```

```
        arr[i+1]=arr[i];
    }
     n++;
    arr[pos-1]=value;
    printf("Inserted Array:");
    for(i=0;i<n;i++)
    {
        printf("%d ",arr[i]);
    }
    return 0;
}
```

OUTPUT:

Enter the number of elements in the array:5

Enter the value to be inserted:89

Enter the position to insert:5

Enter the array elements:3

4

6

7

8

Inserted Array:3 4 6 7 89 8

7.MATRIX MULTIPLICATION

```
#include<stdio.h>
int main()
{
    int a[10][10],b[10][10],c[10][10];
    int i,k,j,r1,r2,c1,c2;
    printf("Enter the number of rows and columns in matrix1:");
    scanf("%d %d",&r1,&c1);
```

```
printf("Enter the number of rows and columns in matrix2:");
scanf("%d %d",&r2,&c2);
if(c2!=r2)
{
    printf("Matrix multiplication is not possible\n");
    return 0;
}
printf("Enter values of matrix1:");
for(i=0;i<r1;i++)
{
    for(j=0;j<c1;j++)
    {
        scanf("%d",&a[i][j]);
    }
}
printf("Enter values of matrix2:");
for(i=0;i<r2;i++)
{
    for(j=0;j<c2;j++)
    {
        scanf("%d",&b[i][j]);
    }
}
for(i=0;i<r1;i++)
{
    for(j=0;j<c2;j++)
    {
        c[i][j]=0;
        for(k=0;k<c1;k++)
```

```
            {
                c[i][j]+=a[i][k]*b[k][j];
            }
        }
    }
    printf("Multiplied Matrix:\n");
    for(i=0;i<c1;i++)
    {
        for(j=0;j<r2;j++)
        {
            printf("%d ",c[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
Enter the number of rows and columns in matrix1:3
3
Enter the number of rows and columns in matrix2:3
3
Enter values of matrix1:1
2
3
4
5
6
1
2
```

3

Enter values of matrix2:5

5

6

6

4

3

2

1

2

Multiplied Matrix:

23 16 18

62 46 51

23 16 18

8.MATRIX TRANSPOSE

```
#include<stdio.h>
int main()
{
    int a[10][10],t[10][10],r,c,i,j;
    printf("Enter the number of rows and columns in matrix a:");
    scanf("%d %d",&r,&c);
    printf("Enter values of matrix a:");
    for(i=0;i<r;i++)
    {
        for(j=0;j<c;j++)
        {
            scanf("%d",&a[i][j]);
            t[j][i]=a[i][j];
        }
```

```
    }
    printf("Matrix a:\n");
    for(i=0;i<r;i++)
    {
        for(j=0;j<c;j++)
        {
            printf("%d ",a[i][j]);
        }
        printf("\n");
    }

    printf("Transpose of matrix a:\n");
    for(i=0;i<c;i++)
    {
        for(j=0;j<r;j++)
        {
            printf("%d ",t[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

Enter the number of rows and columns in matrix a:3

3

Enter values of matrix a:1

2

3

3

2

1

4

3

1

Matrix a:

1 2 3

3 2 1

4 3 1

Transpose of matrix a:

1 3 4

2 2 3

3 1 1

9.RIGHT DIAGONAL SUM

```
#include<stdio.h>
int main()
{
    int n,a[10][10],rd_sum=0;
    int i,j;
    printf("Enter the order of matrix:");
    scanf("%d",&n);

    printf("Enter matrix a:");
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
```

```
            scanf("%d",&a[i][j]);
        }
    }

    for(i=0;i<n;i++)
    {
        rd_sum += a[i][n-1-i];
    }

    printf("Right diagonal sum:%d",rd_sum);
    return 0;
}
```

OUTPUT:

```
Enter the order of matrix:3
Enter matrix a:3
3
3
2
2
2
1
1
4
Right diagonal sum:6
```

10.CHECK FOR IDENTITY

```
#include<stdio.h>
int main()
{
    int a[10][10],n,i,j;
```

```
int flag=1;
printf("Enter the order of matrix a:");
scanf("%d",&n);
printf("Enter the values of matrix a:");
for(i=0;i<n;i++)
{
    for(j=0;j<n;j++)
    {
        scanf("%d",&a[i][j]);
    }
}
for(i=0;i<n;i++)
{
    for(j=0;j<n;j++)
    {
        if(i==j)
        {
            if(a[i][j]!=1)
            {
                flag=0;
            }
        }
        else
        {
            if(a[i][j]!=0)
            {
                flag=0;
            }
        }
```

```
        }

    }
    if(flag==0)
    {
        printf("Given matrix isn't identity matrix\n");
    }
    else
    {
        printf("Given matrix is a identity matrix");
    }
    return 0;
}
```

OUTPUT:

Enter the order of matrix a:2

Enter the values of matrix a:1

2

3

4

Given matrix isn't identity matrix

11.TO FIND THE NUMBER OF VOWEL,CONSONANT,DIGITS,WHITE SPACE

```
#include<stdio.h>
int main()
{
    char str[100];
    int vowel=0,consonant=0,digit=0,space=0,i;
    printf("Enter the text:");
    scanf("%[^\n]",str);
```

```
    for(i=0;str[i]!='\0';i++)
    {
        if((str[i]>='A' && str[i]<='Z') || (str[i]>='a' && str[i]<='z'))
        {
            if(str[i]=='A' ||str[i]=='E' || str[i]=='I' || str[i]=='O' || str[i]=='U' || str[i]=='a' ||
str[i]=='e' || str[i]=='i' || str[i]=='o' || str[i]=='u')
            {
                vowel++;
            }
            else
            {
                consonant++;
            }
        }
        else if(str[i]>='0' && str[i]<='9')
        {
            digit++;
        }
        else if(str[i]==' ')
        {
            space++;
        }
    }
    printf("Vowels=%d\n",vowel);
    printf("consonants=%d\n",consonant);
    printf("digits=%d\n",digit);
    printf("White space=%d\n",space);
    return 0;
}
```

OUTPUT:

Enter the text:how beautiful it is!

Vowels=8

consonants=8

digits=0

White space=3

12.FREQUENCY OF A CHAR IN A STRING

```
#include<stdio.h>
int main()
{
   char str[100];
   char ch;
   int i,count=0;
   printf("Enter a text:");
   scanf("%[^\n]",str);
   printf("Enter the character to find:");
   scanf(" %c",&ch);
   for(i=0;str[i]!='\0';i++)
   {
      if(str[i]==ch)
      {
         count++;
      }
   }
   printf("Frequency of the character %c:%d",ch,count);
   return 0;
}
```

OUTPUT:

Enter a text: hello world

Enter the character to find:w

Frequency of the character w:1

13. COUNT WORDS IN A STRING

```
#include <stdio.h>
int main()
{
    char str[100];
    int i,wcount=1,ncount=0,scount=0;
    printf("Enter a string:");
    scanf("%[^\n]",str);
    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]==' ' || str[i]=='\t')
        {
            wcount++;
        }
        else if(str[i]>='0' && str[i]<'9')
        {
            ncount++;
        }
        else
        {
            scount++;
        }
    }
    printf("Number of words in the given string is %d",wcount);
    return 0;
}
```

OUTPUT:

Enter a string:HELLO WORLD

Number of words in the given string is 2

14.SORT STRING ARRAY

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str[10][30],temp[30];
    int i,j,n;
    printf("Enter the number of strings:");
    scanf("%d",&n);
    printf("Enter strings:");
    for(i=0;i<n;i++)
    {
        scanf("%s",str[i]);
    }
    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(strcmp(str[i],str[j])>0)
            {
                strcpy(temp,str[i]);
                strcpy(str[i],str[j]);
                strcpy(str[j],temp);
            }
        }
    }
```

```
    printf("Strings in alphabetical order:");
    for(i=0;i<n;i++)
    {
        printf("%s\n",str[i]);
    }
    return 0;
}
```

OUTPUT:

Enter the number of strings:3

Enter strings:lotus

daisy

tulip

Strings in alphabetical order:daisy

lotus

tulip

15. TOGGLE CASE OF A SENTENCE (LOWERCASE TO CUPPERCASE AND VICEVERSA)

```
#include<stdio.h>
int main()
{
    int i;
    char na[50];
    printf("Enter any string:");
    scanf("%[^\n]",na);
    for(i=0;na[i]!='\0';i++)
    {
        if(na[i]>='a' && na[i]<='z')
        {
```

```
        na[i]=na[i]-32;
      }
      else if(na[i]>='A' && na[i]<='Z')
      {
        na[i]=na[i]+32;
      }
    }
    printf("Toggled case:%s",na);
    return 0;
}
```

OUTPUT:

Enter any string:GOOD morning

Toggled case:good MORNING